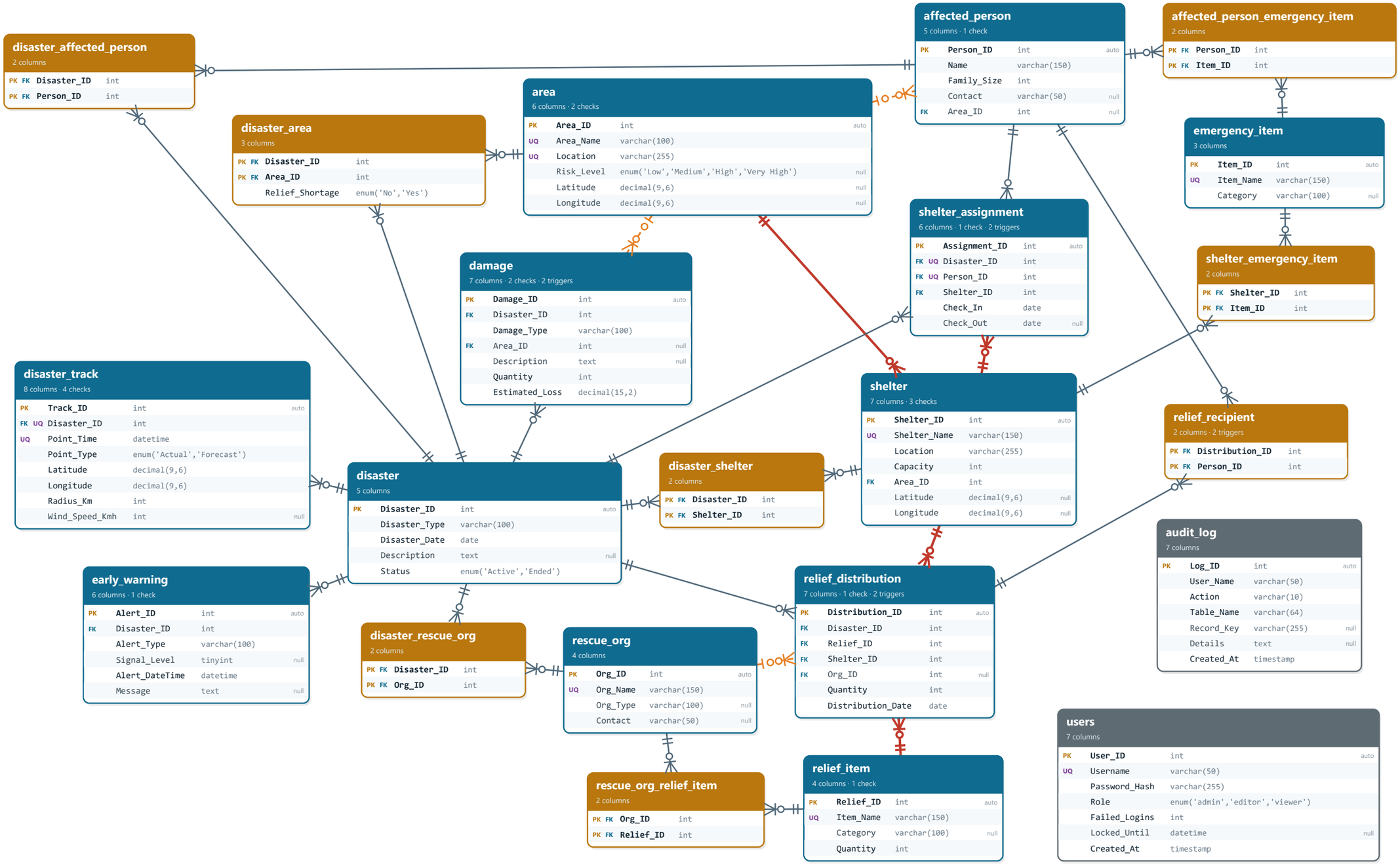
Disaster Mapping & Relief Management System - Entity-Relationship Diagram

Database disaster_and_relief_management (MySQL 8, InnoDB): 22 tables, 99 columns, 29 foreign keys. Generated from MySQL's information_schema and cross-checked against disaster_and_relief_management.sql.



How to read this diagram

- main table
- relationship (link) table: primary key made of foreign keys
- stand-alone table (no foreign keys): users, audit_log

PK primary key **FK** foreign key **UQ** part of a unique key null column allows NULL auto AUTO_INCREMENT

Columns are in table order with their exact MySQL types; bold = primary key. Page 2 lists the 29 foreign keys, 8 unique keys, 16 CHECK constraints and 8 triggers.

Every line is one FOREIGN KEY constraint: it runs from the referenced (parent) table to the table that holds the foreign key (child).

ON DELETE rule: CASCADE (22): child rows are deleted too · SET NULL (3): the foreign key is emptied · RESTRICT (4): deletion is refused

Line ends: || exactly one parent (foreign key is NOT NULL) · |o zero or one parent (foreign key allows NULL) · >o zero or many child rows

Specification behind the diagram

Everything on this page and in the diagram was read from MySQL (information_schema) and compared with the install file disaster_and_relief_management.sql: tables, columns, keys, constraints and triggers all match.

Foreign keys (29)

#	Parent (referenced)	Child (holds the foreign key)	Parent side	Child side	ON DELETE	ON UPDATE	Constraint name
1	**affected_person**.Person_ID	**affected_person_emergency_item**.Person_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_person_item_person`
2	**affected_person**.Person_ID	**disaster_affected_person**.Person_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_disaster_person_person`
3	**affected_person**.Person_ID	**relief_recipient**.Person_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_recipient_person`
4	**affected_person**.Person_ID	**shelter_assignment**.Person_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_assignment_person`
5	**area**.Area_ID	**affected_person**.Area_ID	zero or one parent	zero or many	SET NULL	CASCADE	`fk_person_area`
6	**area**.Area_ID	**damage**.Area_ID	zero or one parent	zero or many	SET NULL	CASCADE	`fk_damage_area`
7	**area**.Area_ID	**disaster_area**.Area_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_disaster_area_area`
8	**area**.Area_ID	**shelter**.Area_ID	exactly one parent	zero or many	RESTRICT	CASCADE	`fk_shelter_area`
9	**disaster**.Disaster_ID	**damage**.Disaster_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_damage_disaster`
10	**disaster**.Disaster_ID	**disaster_affected_person**.Disaster_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_disaster_person_disaster`
11	**disaster**.Disaster_ID	**disaster_area**.Disaster_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_disaster_area_disaster`
12	**disaster**.Disaster_ID	**disaster_rescue_org**.Disaster_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_disaster_rescue_disaster`
13	**disaster**.Disaster_ID	**disaster_shelter**.Disaster_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_disaster_shelter_disaster`
14	**disaster**.Disaster_ID	**disaster_track**.Disaster_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_track_disaster`
15	**disaster**.Disaster_ID	**early_warning**.Disaster_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_early_warning_disaster`
16	**disaster**.Disaster_ID	**relief_distribution**.Disaster_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_distribution_disaster`
17	**disaster**.Disaster_ID	**shelter_assignment**.Disaster_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_assignment_disaster`
18	**emergency_item**.Item_ID	**affected_person_emergency_item**.Item_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_person_item_item`
19	**emergency_item**.Item_ID	**shelter_emergency_item**.Item_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_shelter_item_item`
20	**relief_distribution**.Distribution_ID	**relief_recipient**.Distribution_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_recipient_distribution`
21	**relief_item**.Relief_ID	**relief_distribution**.Relief_ID	exactly one parent	zero or many	RESTRICT	CASCADE	`fk_distribution_relief`
22	**relief_item**.Relief_ID	**rescue_org_relief_item**.Relief_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_org_relief_item`
23	**rescue_org**.Org_ID	**disaster_rescue_org**.Org_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_disaster_rescue_org`
24	**rescue_org**.Org_ID	**relief_distribution**.Org_ID	zero or one parent	zero or many	SET NULL	CASCADE	`fk_distribution_org`
25	**rescue_org**.Org_ID	**rescue_org_relief_item**.Org_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_org_relief_org`
26	**shelter**.Shelter_ID	**disaster_shelter**.Shelter_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_disaster_shelter_shelter`
27	**shelter**.Shelter_ID	**relief_distribution**.Shelter_ID	exactly one parent	zero or many	RESTRICT	CASCADE	`fk_distribution_shelter`
28	**shelter**.Shelter_ID	**shelter_assignment**.Shelter_ID	exactly one parent	zero or many	RESTRICT	CASCADE	`fk_assignment_shelter`
29	**shelter**.Shelter_ID	**shelter_emergency_item**.Shelter_ID	exactly one parent	zero or many	CASCADE	CASCADE	`fk_shelter_item_shelter`

Tables (22)

Table	Kind	Cols	Primary key	Foreign key columns	Checks	Triggers
affected_person	main	5	Person_ID	Area_ID	1	0
affected_person_emergency_item	relationship	2	Person_ID, Item_ID	Item_ID, Person_ID	0	0
area	main	6	Area_ID	-	2	0
audit_log	stand-alone	7	Log_ID	-	0	0
damage	main	7	Damage_ID	Area_ID, Disaster_ID	2	2
disaster	main	5	Disaster_ID	-	0	0
disaster_affected_person	relationship	2	Disaster_ID, Person_ID	Disaster_ID, Person_ID	0	0
disaster_area	relationship	3	Disaster_ID, Area_ID	Area_ID, Disaster_ID	0	0
disaster_rescue_org	relationship	2	Disaster_ID, Org_ID	Disaster_ID, Org_ID	0	0
disaster_shelter	relationship	2	Disaster_ID, Shelter_ID	Disaster_ID, Shelter_ID	0	0
disaster_track	main	8	Track_ID	Disaster_ID	4	0
early_warning	main	6	Alert_ID	Disaster_ID	1	0
emergency_item	main	3	Item_ID	-	0	0
relief_distribution	main	7	Distribution_ID	Disaster_ID, Org_ID, Relief_ID, Shelter_ID	1	2
relief_item	main	4	Relief_ID	-	1	0

Unique keys (8)

Table	Key	Columns
area	`uq_area_name_location`	Area_Name, Location
disaster_track	`uq_track_disaster_time`	Disaster_ID, Point_Time
emergency_item	`uq_emergency_item_name`	Item_Name
relief_item	`uq_relief_item_name`	Item_Name
rescue_org	`uq_rescue_org_name`	Org_Name
shelter	`uq_shelter_name`	Shelter_Name
shelter_assignment	`uq_assignment_disaster_person`	Disaster_ID, Person_ID
users	`uq_users_username`	Username

CHECK constraints (16)

Table	Constraint	Rule
affected_person	`chk_person_family`	(`Family_Size` > 0)
area	`chk_area_lat`	(`Latitude` between -(90) and 90)
area	`chk_area_lng`	(`Longitude` between -(180) and 180)
damage	`chk_damage_loss`	(`Estimated_Loss` >= 0)

relief_recipient	relationship	2	Distribution_ID, Person_ID	Distribution_ID, Person_ID	0	2
rescue_org	main	4	Org_ID	-	0	0
rescue_org_relief_item	relationship	2	Org_ID, Relief_ID	Relief_ID, Org_ID	0	0
shelter	main	7	Shelter_ID	Area_ID	3	0
shelter_assignment	main	6	Assignment_ID	Disaster_ID, Person_ID, Shelter_ID	1	2
shelter_emergency_item	relationship	2	Shelter_ID, Item_ID	Item_ID, Shelter_ID	0	0
users	stand-alone	7	User_ID	-	0	0

damage	chk_damage_qty	(`Quantity` >= 0)
disaster_track	chk_track_lat	(`Latitude` between -(90) and 90)
disaster_track	chk_track_lng	(`Longitude` between -(180) and 180)
disaster_track	chk_track_radius	(`Radius_Km` > 0)
disaster_track	chk_track_wind	(`Wind_Speed_Kmh` >= 0)
early_warning	chk_warning_signal	(`Signal_Level` between 1 and 10)
relief_distribution	chk_distribution_qty	(`Quantity` > 0)
relief_item	chk_relief_qty	(`Quantity` >= 0)
shelter	chk_shelter_capacity	(`Capacity` > 0)
shelter	chk_shelter_lat	(`Latitude` between -(90) and 90)
shelter	chk_shelter_lng	(`Longitude` between -(180) and 180)
shelter_assignment	chk_assignment_dates	((`Check_Out` is null) or (`Check_Out` >= `Check_In`))

Triggers (8)

Trigger	Table	When
trg_damage_ins	**damage**	BEFORE INSERT
trg_damage_upd	**damage**	BEFORE UPDATE
trg_distribution_ins	**relief_distribution**	BEFORE INSERT
trg_distribution_upd	**relief_distribution**	BEFORE UPDATE
trg_recipient_ins	**relief_recipient**	BEFORE INSERT
trg_recipient_upd	**relief_recipient**	BEFORE UPDATE
trg_assignment_ins	**shelter_assignment**	BEFORE INSERT
trg_assignment_upd	**shelter_assignment**	BEFORE UPDATE

Triggers refuse a stay, distribution, recipient or damage record whose shelter, family, organisation or area does not belong to that disaster. Capacity and stock limits are checked by the application (inc/rules.php) inside a transaction with row locks.